

Mass distribution principle (Chapter 4.1 in Falconer)

Defn: A mass distribution on $\mathbb{R}^n$ is a measure μ s.t. $0 < \mu(\mathbb{R}^n) < \infty$.

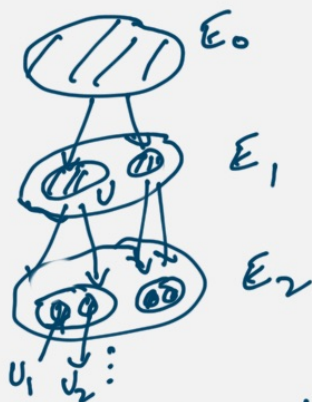
Examples:

① Point mass distribution:
Let $a \in \mathbb{R}^n$. Define for $A \subseteq \mathbb{R}^n$

$$\mu(A) = \begin{cases} 1 & \text{if } a \in A \\ 0 & \text{if } a \notin A \end{cases}$$

② Uniform mass distribution on $[0,1]$:

$$\mu(A) = \mathcal{L}^1(A \cap [0,1]) \text{ for } A \subseteq \mathbb{R}$$



$$\mu(U_1) + \mu(U_2) = \mu(U)$$

For any $A \subseteq \mathbb{R}^n$ define

$$\mu(A) = \inf \left\{ \sum_{i=1}^{\infty} \mu(U_i) : A \cap E_0 \subseteq \bigcup_{i=1}^{\infty} U_i \text{ and } U_i \in \mathcal{E} \right\}$$

$$E_{\infty} = \bigcap_{n=1}^{\infty} E_n$$

$$\mu(E_0) = 1$$

$$\mu(E_n) = 1 \quad \forall n$$

$$\mu(E_{\infty}) = 1$$

It can be shown μ defines an outer measure on $\mathbb{R}^n$ which is an extension of μ on Σ .

Also, $\text{support}(\mu) = E_\infty$.

Theorem (Mass distribution principle):

Let μ be a mass distribution on a subset F of $\mathbb{R}^n$ and suppose for some $s > 0$,
 $\exists c > 0$ and $\delta_0 > 0$ s.t.
 $\mu(U) \leq c |U|^s \quad \forall U \text{ with } |U| \leq \delta_0$.

Then $\mathcal{H}^s(F) \geq \frac{\mu(F)}{c}$ and $\dim_H F \geq s$.

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Proof: Let $\delta \leq \delta_0$. If $\{U_i\}_{i=1}^\infty$ is a δ -cover of F , then

$$\begin{aligned} \mu(F) &\leq \mu\left(\bigcup_{i=1}^\infty U_i\right) \leq \sum_{i=1}^\infty \mu(U_i) \\ &\leq c \sum_{i=1}^\infty |U_i|^s \end{aligned}$$

Taking infimum over all δ -covers of F , we get

$$\mu(F) \leq c \mathcal{H}_\delta^s(F)$$

$$\Rightarrow \mathcal{H}_\delta^s(F) \geq \frac{\mu(F)}{c} \quad \forall \delta \leq \delta_0$$

$$\Rightarrow \mathcal{H}^s(F) \geq \frac{\mu(F)}{c} > 0$$

Since $\mathcal{H}^s(F) > 0$, $\dim_H F \geq s$

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Example: Cantor set:

We can define a mass distribution on the Cantor set F such that the measure of each 2^k intervals at level k is $\frac{1}{2^k}$.

Let U be any set with $|U| < 1$ and let k be an integer s.t. $\frac{1}{3^k} \leq |U| < \frac{1}{3^{k-1}}$.

Then U can intersect at most one of the 2^k intervals at level k .

$$\therefore \mu(U) \leq \frac{1}{2^k} = \left(\frac{\ln 2}{\ln 3} \right)^{-k} \\ = \left(\frac{1}{3^k} \right)^{\frac{\ln 2}{\ln 3}} \leq (3|U|)^{\frac{\ln 2}{\ln 3}} = 2|U|^{\frac{\ln 2}{\ln 3}}$$

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$\therefore$ By the mass distribution principle,

$$\mu^{\frac{\ln 2}{\ln 3}}(F) \geq \frac{\mu(F)}{2} = \frac{1}{2}$$

$$\text{and } \dim_H F \geq \frac{\ln 2}{\ln 3}.$$

Theorem: Suppose $\{\mathbb{R}^n; f_1, f_2, \dots, f_N\}$ be an IFS with contractions $f_1, \dots, f_N$ such that

$$\|f_i(x) - f_i(y)\| \leq b_i \|x - y\| \quad \forall x, y,$$

where $0 < b_i < 1$ for each i .

Suppose the attractor F is such that $F = \bigcup_{i=1}^N f_i(F)$ and $f_i(F) \cap f_j(F) = \emptyset$ for $i \neq j$.

Then $\dim_H F \geq s$, where $\sum_{i=1}^N b_i^s = 1$.

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